SRM

DEPARTMENT OF CHEMISTRY

College of Engineering and Technology
SRM Institute of Science and Technology
Kattankulathur – 603 203

SET – 1

INTERNAL ASSESSMENT – I (CLA1-T1)

Program: B.Tech	Date: 11-09-2023
Course Code & Title: 21CYB101J & Chemistry	Time: 8.00-8.50am
Year & Sem: I Year & I Sem	Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. The crystal field splitting energy of an octahedral complex is 18,000 cm^{-1}. What will be the energy in terms of Δ_t ?
   a) 40,500 cm^{-1} b) 8100 cm^{-1} c) 36,000 cm^{-1} d) 9000 cm^{-1}

2. The number of unpaired electrons present in $[Fe(CN)_6]^{4-}$ is
   a) 0 b) 1 c) 2 d) 3

3. Arrange the following ligands, en, NH_3, H_2O and CN^- in their decreasing order of field strength
   a) $NH_3 > CN^- > H_2O >$ en
   b) en $> CN^- > H_2O > NH_3$
   c) $CN^- >$ en $> NH_3 > H_2O$
   d) $CN^- >$ en $> H_2O > NH_3$

4. The complexes $[Co(NH_3)_6][Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6][Co(C_2O_4)_3]$
   (a) Linkage isomerism
   (b) Geometrical isomerism
   (c) Coordination isomerism
   (d) Ionisation isomerism

5. The shielding constant of 1s electron in helium atom is
   a) 1.70 b) 0.35 c) 1.65 d) 0.30

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Part – B (2 x 10 = 20 Marks)

6. a. (i) Discuss the salient features of crystal field theory. *(4 Marks)*

(ii) Identify the complex possessing the larger value of Δ_o in the following pairs and give reason. *(6 Marks)*

(A) $[Ni(CN)_6]^{4-}$ and $[Ni(H_2O)_6]^{2+}$

(B) $[Co(H_2O)_6]^{2+}$ and $[Co(H_2O)_6]^{3+}$

(C) $[Rh(NH_3)_6]^{3+}$ and $[Ir(NH_3)_6]^{3+}$

(OR)

b. Discuss the crystal field splitting of a tetrahedral complex. Calculate the CFSE and magnetic moment of $[NiCl_4]^{2-}$ (Atomic number of Ni is 28). *(10 Marks)*

7. a. Brief about the following isomerism exhibited by the coordination compounds with a suitable example for each. *(3+3+4 Marks)*

(i) Ionization

(ii) Hydrate

(iii) Geometrical

(OR)

b. (i) The UV-Visible spectrum of an octahedral complex showed λmax at 600 nm. Calculate the crystal field splitting energy, Δ_o and complementary colour of the complex? *(6 Marks)*

(ii) Determine the effective nuclear charge experienced by the 4p electron of Bromine. [Atomic number of Br = 35]. *(4 Marks)*

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Set-1

INTERNAL ASSESSMENT – 1
ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 11.09.23
Duration: 8.00-8.50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1 The crystal field splitting energy of an octahedral complex is 18,000 cm^{-1}. What will be the energy in terms of Δ_t?

b) 8100 cm^{-1}

2 The number of unpaired electrons present in $[Fe(CN)_6]^{4-}$ is

a) 0

3 Arrange the following ligands. en, NH_3, H_2O and CN^- in their decreasing order of field strength

c) $CN^- > en > NH_3 > H_2O$

4 The complexes $[Co(NH_3)_6][Cr(C_2O_4)_3]$ and $[Cr(NH_3)_6][Co(C_2O_4)_3]$

c) Coordination isomerism

5 The shielding constant of 1s electron in helium atom is_____

d) 0.30

Part – B (2 x 10 = 20 Marks)

6. a. (i) Discuss the salient features of crystal field theory. (4 Marks)

Important features of CFT – 4 Points – 4 Marks

(ii) Identify the complex possessing the larger value of Δ_o in the following pairs and give reason. (6 Marks)

Explanation for the high value of Δ_o in each pair– 2 Marks – (3x2 = 6 Marks)

(OR)

b. Discuss the crystal field splitting of a tetrahedral complex. Calculate the CFSE and magnetic moment of $[NiCl_4]^{2-}$ (Atomic number of Ni is 28). (10 Marks)

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Crystal Field Splitting Diagram- 2 Marks,
Explanation on d – orbitals splitting – 4 Marks,
CFSE calculation – 2 Marks
Magnetic moment calculation – 2 Marks

7. a. Brief about the following isomerism exhibited by the coordination compounds with a suitable example for each. (3+3+4 Marks)

(i) Ionization isomerism - **definition with an example – 3 Marks**

(ii) Hydrate isomerism - **definition with an example – 3 Marks**

(iii) Geometrical isomerism - **definition with an example – 4 Marks**

(OR)

b. (i) The UV-Visible spectrum of an octahedral complex showed λ_{max} at 600 nm.

Calculate the crystal field splitting energy, Δ_o and complementary colour of the complex? (6 Marks)

$\Delta_0 = hc/\lambda = (6.626 \times 10^{-34}\ J.s)(3 \times 10^{8} cm/s)/(600 \times 10^{-9} m) = 3.313 \times 10^{-19}\ J$ – 4 Marks

Wavelength of light absorbed is 600 nm (i.e. orange light) and hence the complementary colour of the complex is blue – 2 Marks

(ii) Determine the effective nuclear charge experienced by the 4p electron of Bromine. [Atomic number of Br = 35]. (4 Marks)

Slater configuration of Br: $(1s^2)\ (2s^2\ 2p^6)\ (3s^2\ 3p^6)\ (3d^{10})\ (4s^2\ 4p^5)$ - 1 Mark

Shielding Constant of 4p=[(10x1)+(18x0.85)+(6x0.35)] = 27.4 – 2 Marks

Effective nuclear charge = 35 – 27.4 = 7.60 – 1 Mark

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SET – 2

INTERNAL ASSESSMENT – I (CLA1-T1)

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & I Sem

Date: 11-09-2023
Duration: 8.00-8.50am
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1. Among the following complexes, the one which shows zero crystal field stabilization energy (CFSE) is
   a) $[Fe(H_2O)_6]^{3+}$ b) $[Mn(H_2O)_6]^{3+}$ c) $[Co(H_2O)_6]^{3+}$ d) $[Co(H_2O)_6]^{2+}$

2. The complex possessing high magnetic moment value is _____
   (a) $[Cr(H_2O)_6]^{3+}$ (b) $[Fe(H_2O)_6]^{2+}$
   (c) $[Cu(H_2O)_6]^{2+}$ (d) $[Zn(H_2O)_6]^{2+}$

3. Which one of the following complexes is not coloured?
   a) $[Cu(Cl_4)]^{2-}$ b) $[Cd(Cl_4)]^{2-}$ c) $[Fe(Cl_6)]^{4-}$ d) $[Fe(CN)_6]^{4-}$

4. The type of isomerism shown by $[CoCl_2(en)_2]$ is
   a) Ionization b) Geometrical c) Linkage d) Coordination

5. The effective nuclear charge of 2s electron in Beryllium atom is _____
   a) 2.05 b) 1.95 c) 0.70 d) 3.30

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Part – B (2 x 10 = 20 Marks)

6. a. Describe the crystal field splitting of an octahedral complex with neat diagram. Show the CFSE and magnetic moment calculations of both low spin and high spin complexes for the metal ion with d^7 electron configuration. (10 Marks)

(OR)

b. (i) Write a short note on Spectrochemical series. (4 Marks)

(ii) Compare the CFSE and magnetic moment values of $[MnCl_4]^{2-}$ and $[CoCl_4]^{2-}$ and $[CuCl_4]^{2-}$ complexes. [Atomic numbers of Mn, Co and Cu are 25, 27 and 29 respectively]. (6 Marks)

7. a. Discuss the following isomerism shown by the coordination compounds with a suitable example for each. (3+3+4 Marks)

(i) Linkage

(ii) Coordination

(iii) Optical

(OR)

b. (i) Arrange the following complexes in their increasing order of wavelength of light absorbed and explain. (6 Marks)

$[CrF_6]^{3-}$, $[Cr(CN)_6]^{3-}$ and $[Cr(H_2O)_6]^{3+}$

(ii) Calculate the effective nuclear charge experienced by the 3d electron of Zn [Z=30]. (4 Marks)

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Set-2

INTERNAL ASSESSMENT – I
ANSWER KEY

Program: B.Tech
Course Code & Title: 21CYB101J & Chemistry
Year & Sem: I Year & II Sem

Date: 11.09.23
Duration: 8.00-8.50 AM
Max. Marks: 25 marks

Part – A (5 x 1 = 5 Marks)
Answer ALL the Questions

1 Among the following complexes, the one which shows zero crystal field stabilization energy (CFSE) is
a) $[Fe(H_2O)_6]^{3+}$

2 The complex possessing high magnetic moment value is _____
b) $[Fe(H_2O)_6]^{2+}$

3 Which one of the following complexes is not coloured?
b) $[Cd(Cl_4)]^{2-}$

4 The type of isomerism shown by $[CoCl_2(en)_2]$ is
b) Geometrical

5 The effective nuclear charge of 2s electron in Beryllium atom is _____
b) 1.95

Part – B (2 x 10 = 20 Marks)

6. a. Describe the crystal field splitting of an octahedral complex with neat diagram. Show the CFSE and magnetic moment calculations of both low spin and high spin complexes for the metal ion with d^7 electron configuration. (10 Marks)

Crystal Field Splitting Diagram- 2 Marks,
Explanation on d – orbitals splitting – 4 Marks,
CFSE calculations (both high & low spin) – 2 Marks
Magnetic moment calculations (both high & low spin) – 2 Marks

(OR)

b. (i) Write a short note on Spectrochemical series. (4 Marks)

Definition – 2 Marks
Arrangement of ligands in series – 2 Marks

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(ii) Compare the CFSE and magnetic moment values of $[MnCl_4]^{2-}$ and $[CoCl_4]^{2-}$ and $[CuCl_4]^{2-}$ complexes. [Atomic numbers of Mn, Co and Cu are 25, 27 and 29 respectively]. (6 Marks)

For each complex,
CFSE calculation – 1 Mark; Magnetic moment calculation – 1 Mark.

7. a. Discuss the following isomerism shown by the coordination compounds with a suitable example for each. (3+3+4 Marks)

(i) Linkage – **definition with an example – 3 Marks**
(ii) Coordination – **definition with an example – 3 Marks**
(iii) Optical – **definition with an example – 4 Marks**

(OR)

b. (i) Arrange the following complexes in their increasing order of wavelength of light absorbed and explain. (6 Marks)

$[Cr(CN)_6]^{3-} < [Cr(H_2O)_6]^{3+} < [CrF_6]^{3-}$ - 2 Marks
Explanation – 4 Marks

(ii) Calculate the effective nuclear charge experienced by the 3d electron of Zn [Z=30]. (4 Marks)

Slater configuration of Zn: $(1s^2)$ $(2s^2\ 2p^6)$ $(3s^2\ 3p^6)$ $(3d^{10})$ $(4s^2)$ - 1 Mark

Screening constant = [(18x1) + (9x0.35)] = 21.15 - 2 Marks

Effective nuclear charge = 30 – 21.15 = 8.85 - 1 Mark

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